

Roll No.

TPE-503

B. TECH. (PE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
PETROLEUM PRODUCTION ENGINEERING-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Describe Surface and Subsurface Production Equipment.

10 Marks (CO1, CO2)

OR

(b) With a neat and clean diagram explain the concept of vertical and horizontal two-phase separators.

10 Marks (CO1, CO2)

2. (a) With the help of a figure explain function of oil and gas facility.

10 Marks (CO1, CO2)

OR

(b) (i) With the help of a table of weight percentage explain composition of oil and gas.

(ii) Describe Bridge Plugs, Safety Valves and Packers.

10 Marks (CO1, CO2)

P. T. O.

(2)

3. (a) Explain Emulsion and types of emulsion. 10 Marks (CO2)

OR

- (b) Describe all physical and chemical properties of crude oil.

10 Marks (CO2)

4. (a) Describe gas sweetening, types of amines used for gas treating, process of gas treating.

10 Marks (CO2)

OR

- (b) Briefly explain the following :

10 Marks (CO2)

(i) Paraffins

(ii) Aromatics

(iii) Naphthenes

(iv) Branched structure of C_4H_{10} , C_6H_{14} and also write their boiling point

5. (a) Write short notes on the following :

10 Marks (CO1, CO2)

(i) Well Flowline

(ii) Gathering Pipeline

(iii) GGS

(iv) Transmission Pipeline

OR

- (b) Describe Flocculation or Aggregation and Coalescence.

10 Marks (CO1, CO2)